

Transport Intelligence Report

Busiest / Low-Performing / Overcrowded / Underutilized Routes

All routes have a 0-100 performance index (utilization 25%, punctuality 20%, delay 15%, congestion 15%, demand 10%, delay rate 8% and travel time 7%) in addition to a behavioral index:

Classification	Routes
High Performing	42
High Demand, Unreliable	24
Reliable, Underutilized	19
Low Performing	10
Overcrowded	5

Top-scoring route: R016 (75.1). Lowest-scoring: R006 (42.3).

Peak Periods

Based on actual boarding times rather than predetermined time slots. Hour with most departures is 08:00 and 4,670,516 people boarded. Number of boardings per day type: weekdays 21,111,388 (average of 842.8 per route-day), weekends 3,636,971 (363.0), holidays 171,625 (218.4).

Most Delayed / Most Reliable Routes

Best on-time record (highest percentage of on-time departures, for trips only): R030, 60.1%. Most frequent delay location: route R021 at 17:00, average delay of 16.3 minutes on 278 trips , the top entry in the table with the most recurring route-hour pattern (at least 40 trips needed for pattern formation).

Passenger-Flow Patterns

Boarding / alighting counted per stop on five time-of-day bands. Busiest stop in terms of turnover: S0141 (15,809 boardings, 11,961 alightings, net accumulation of +3,848 – net origin). The demand is clearly asymmetrical in terms of direction: e.g. route R066 has 311,003 trips in direction 1 compared to 310,924 in direction 0.

Bottleneck Stops

Identified based on the behaviour (30% delay, 22% crowding, 14% accumulation, 14% turnover, 12% peaking, 8% connectivity), filtered at 97th percentile of the score distribution. Top bottleneck: S0141 (score of 0.661, average delay of 7.23 minutes, 24% of trips through it are overcrowded, serves 8 routes). 19 stops identified (top 3%).

Demand Forecasts and Crowding Risk

14-day ahead forecast MAE 147.15 (178.25 baseline). 22,416 trips in the holdout window have the overcrowding probability of 0.75 or higher (compared to 0.5 decision threshold used in the classification metrics).

Route Clusters

K-Means, $k=6$, silhouette = 0.4028. Classified using standardised average daily demand, average occupancy, average delay, reliability and number of overcrowded trips. This mediocre silhouette is truthfully provided in soft ordering rather than hard classification, since the routes have been generated from a continuous set of parameters without any inherent gap in clusters.

Service Gaps

Frequency judgements: 71 good, 20 excessive, 9 insufficient frequency. Supply-demand judgement: 56 balanced, 32 demand excess, 12 supply excess (ranges: 90%+ of supply means demand excess, $\leq 35\%$ of supply means supply excess).

Schedule Recommendations

153 priority recommendations have been made, with evidence for each: 24 critical, 80 high, 30 medium, 19 low.